

Perioperative management of anticoagulant and antiplatelet

Prepared by: PRIC

Disclaimer:

The recommendations listed in this document is intended for educational & informational purposes. It is NOT intended to take the place of physician judgement in diagnosing and treatment of particular patients.

NOAC

Non-cardiac surgery

NOAC [1], [2], [3], [4], [11]	Estimated drug half life (hr)	CrCl (ml/min)	Interval between last dose and surgery (hr)		Resumption after surgery (hr)		Remark
			Low procedural bleed risk	High procedural bleed risk	Low procedural /patient bleed risk	High procedural /patient bleed risk	
Dabigatran	13	≥80	24	48	24	48 - 72	1. Bridging is not required perioperatively except patient with high thromboembolism risk. 2. Only resume NOAC after complete hemostasis achieved over procedural site.
	15	50-79	24-36	48-72			
	18	30-49	48-72	72-96			
Rivaroxaban	6-15	≥30	24	48			
	9	15-29	36	48			
Apixaban	6-15	≥30	24	48			
	17	15-29	36	72-96			

1. Suggest to withhold DOAC 2-3 drug half lives for low procedural bleed risk.
2. Suggest to withhold 4-5 drug half lives for uncertain, intermediate or high procedural bleed risk.
3. Minimal-bleed risk procedure (eg. Minor dental procedure, cataract procedure, minor skin procedure like skin biopsy): Likely safe not to interrupt anticoagulation. If interruption is necessary, consider:
 Dabigatran: (A) Withhold on the day of procedure; (B) Delay AM dose 2-3 hours post-procedure, continue PM as usual if procedure in the morning
 Rivaroxaban: (A) Withhold on the day of procedure; (B) Delay the day dose for 4-6 hours after procedure
 Apixaban: (A) Withhold on the day of procedure; (B) Delay AM dose 2-3 hours post procedure, continue PM as usual if procedure in the morning

NOAC**Cardiac surgery (Major surgery with high bleed risk)**

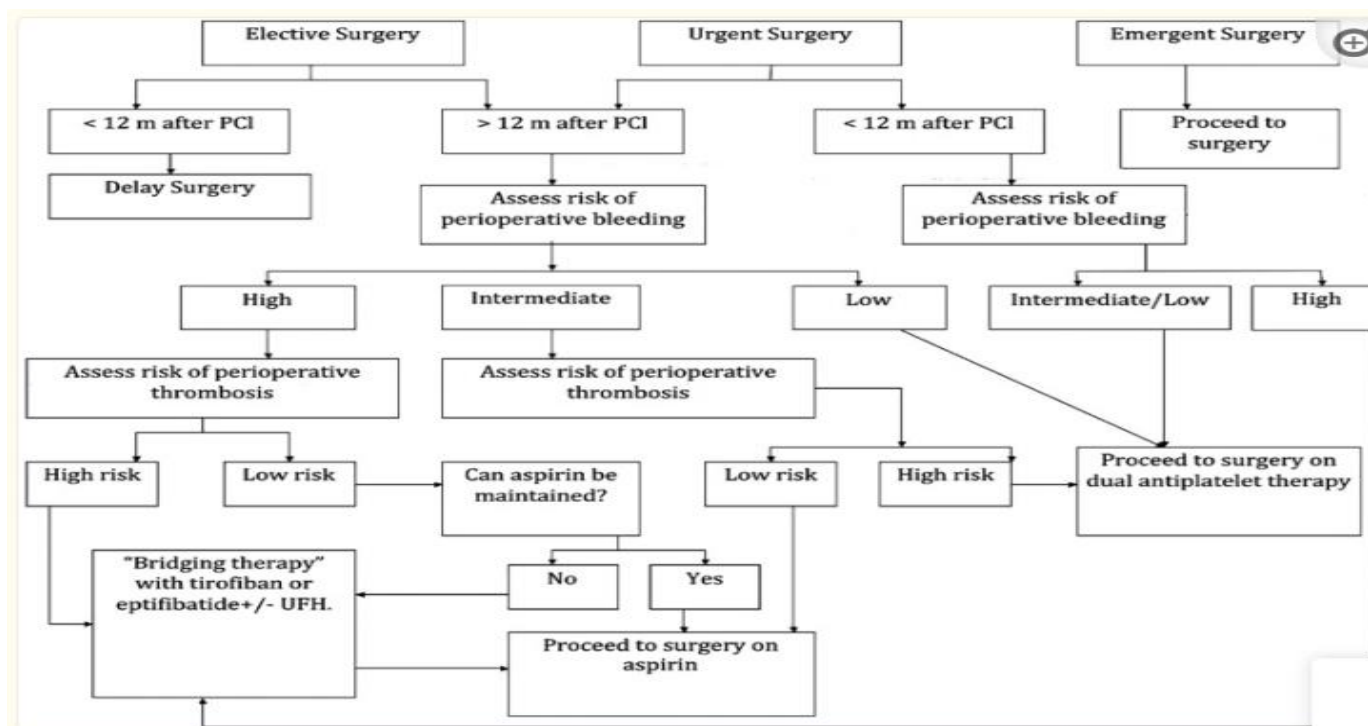
NOAC [11], [14], [16]	Estimated drug half life (hr)	CrCl (ml/min)	Interval between last dose and surgery (hr)	Resumption after surgery (hr)	Remark
Dabigatran	13	≥80	48	72	1. Bridging is not required perioperatively except patient with high thromboembolism risk. 2. Only resume NOAC after complete hemostasis achieved over procedural site.
	15	50-79	72		
	18	30-49	96		
Rivaroxaban	6-15	≥50	48		
		30-49	72		
	9-13	<30	96		
Apixaban	6-15	≥50	48		
		30-49	72		
	17	<30	96		

Parenteral Anticoagulant

Parenteral Anticoagulant	Estimated drug half life (hr)	Interval between last dose and surgery (hr)	Resumption after surgery (hr)		Remark
			Low procedural /patient bleed risk	High procedural /patient bleed risk	
Unfractionated heparin (IVI) [2], [6], [7], [8]	1.5 (anticoagulation effect half life)	4-6	24 (No bolus dose is required)	48-72 (No bolus dose is required)	Monitor coagulation profile after stopped for 6 hours and ensure normal before surgery.
SC Enoxaparin (therapeutic dose – weight based) [2], [6], [7], [8]	4.5-7 HD: 8	24	24	48-72	-
SC Enoxaparin (Prophylaxis/Low dose – non weight based) [7], [8]		12	6		-
Fondaparinux [2], [9]	17-21	36-42	6-12		Limited data. Recommendation made based on neuraxial procedures.

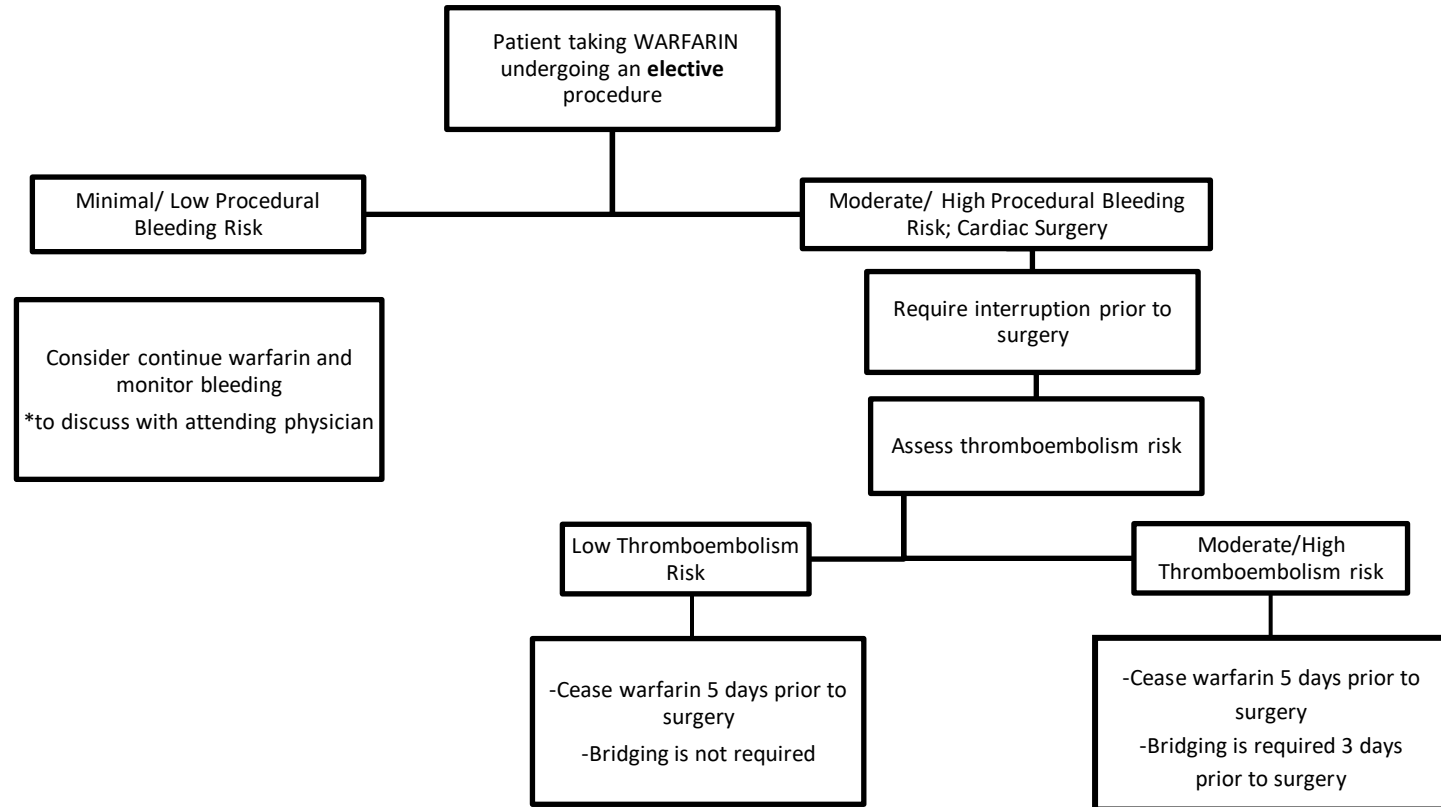
Antiplatelet

Antiplatelet [2], [6], [10], [11], [15], [16]	Interval between last dose and surgery		Resumption after surgery (hr)		Remark
	Minimal-Low procedural/patient bleed risk	Cardiac surgery; High procedural/patient bleed risk; Patient refuse blood transfusion	Low procedural/patient bleed risk	High procedural /patient bleed risk	
Aspirin	Continue	≥5 days CABG: To continue throughout preoperative period DAPT: Continue aspirin if stop P2Y12 inhibitor	Restart as soon as possible		Early withdrawal of antiplatelet therapy for secondary prevention of CVS disease will increase the risk of postoperative myocardial infarction and death 5- to 10-fold.
Clopidogrel		5-7 days	24	24-48	
Ticagrelor		3-5 days			
Ticlopidine		≥14 days			
Prasugrel		≥7 days			



Perioperative management of warfarin [2],[6],[7],[11],[13],[16],[17]

Kindly refer next page for the more detailed interruption of warfarin prior to surgery.



Moderate/High Surg Bleeding Risk
Low Thromboembolic Risk

-Cease warfarin 5 days prior to surgery
-Bridging is not required

	6 days prior to surgery	5 days prior to surgery	4 days prior to surgery	3 days prior to surgery	2 days prior to surgery	1 day prior to surgery	Morning of surgery	Resumption after surgery
Warfarin	Last dose	x	x	x	x	x	x	1. Restart warfarin after 24 hours with bridging if no major bleeding risk when there is adequate haemostasis.
INR						Check if INR < 1.5. Only proceed with surgery when the INR is ≤1.5		2. Delay resumption 48-72hrs post-surgery for major bleeding risk.

Moderate/High Risk Bleeding Risk
Moderate/ High Thromboembolic Risk

-Cease warfarin 5 days prior to surgery
-Bridging is required

	6 days prior to surgery	5 days prior to surgery	4 days prior to surgery	3 days prior to surgery	2 days prior to surgery	1 day prior to surgery	Morning of surgery	Resumption after surgery
Warfarin	Last dose	x	x	x	x	x	x	1. Restart warfarin after 24 hours with bridging if no major bleeding risk when there is adequate haemostasis.
INR						Check if INR < 1.5. Only proceed with surgery when the INR is ≤1.5		2. Delay resumption 48-72hrs post-surgery for major bleeding risk.
Enoxaparin		x		✓	✓	Cease 24 hr before surgery		
IVI Heparin*		x		✓	✓	✓	Cease 4-6 hr before surgery	

*Bridging with an intravenous unfractionated heparin infusion should be reserved for only those patients for whom bridging with Enoxaparin (LMWH) is contraindicated

- If INR is <1.5, a **low** dose of oral [vitamin K](#) (eg, 1 to 2 mg) can be given for selected patients and/or procedures in which a normalized INR is required; this can be followed by re-checking an INR the following day. IV Vitamin K can be given if required faster reversal INR.
- If faster reversal of INR is required, 20-30 IU/kg of 4-factor prothrombin complex concentrates (PCC) is suggested.

Table 1

Suggested risk stratification scheme for perioperative arterial and venous thromboembolism

Thromboembolic risk category	Clinical indication for warfarin therapy		
	Atrial fibrillation	Mechanical heart valves	Venous thromboembolism
High	- CHADS₂ score: 5 or 6 - Recent (< 3 months) stroke/TIA - Rheumatic valvular heart disease	- Any mechanical mitral valve - Older aortic mechanical valve (caged-ball, tilting disk) - Recent (< 3 months) stroke or TIA	- Recent (< 3 months) VTE - Severe thrombophilia*
Moderate	CHADS ₂ score: 3 or 4	Bileaflet aortic valve prosthesis with at least one risk factor†	- VTE within past 3-12 months - Non-severe thrombophilia‡ - Recurrent VTE - Active cancer
Low	CHADS ₂ score: 0-2 (without previous stroke or TIA)	Bileaflet aortic bileaflet without any risk factors†	VTE > 12 months ago

Adapted with permission from Douketis J, et al. *Chest*. 2008;133(6 suppl):299S-339S. ⁷

Procedural Bleeding Risk

Intermediate/ High/ Uncertain bleed risk			
- Complex dental (eg. Extract > 3 teeth) - Cath/ PCI (transfemoral) - Lung bx - Hysterectomy	- Arterial revascularization - Laft atrial appendage occlusion - Lumbar puncture	- Highly vascularized organs (kidney, liver spleen) - Cardiac, intracranial, or spinal - Pericardiocentesis	- Pacemaker/ defib implantation - AF ablation (trans v) - Cervical/ Prostate/ Breast bx - D&C - Cath/ PCI (transradial)
Min bleed risk		Low bleed risk	
- Minor dental - Cataract/ glaucoma - Superficial incisions/excisions - Skin biopsy	- Diagnostic GI endoscopy w or w/o biopsy - Central catheter removal	- Pacemaker/ defib implantation - AF ablation (trans v) - Cervical/ Prostate/ Breast bx - D&C - Cath/ PCI (transradial)	

For full list of procedure, please refer online appendix of 2017 ACC Expert Consensus Decision Pathway for Peri-procedural management.

Reference

1. [Alfonso Tafur, James Douketis](#); Perioperative management of anticoagulant and antiplatelet therapy; Heart. 2018 Sep;104(17):1461-1467. doi: 10.1136/heartjnl-2016-310581. Epub 2017 Dec 7.
2. Javier J. Polania Gutierrez; Klifford R. Rocuts; Perioperative Anticoagulation Management; Treasure Island (FL): [StatPearls Publishing](#); 2023 Jan-
3. J. Wagner, J.F. lock, C. Kastner et al; Perioperative management of anticoagulant therapy; [Innov Surg Sci](#). 2019 Dec; 4(4): 144–151. Published online 2019 Jul 18. doi: [10.1515/iss-2019-0004](#)
4. J. Steffel, P. Verhamme, T.S. Potparra; The 2018 European Heart Rhythm Association Practical Guide on the use of non-vitamin K antagonist oral anticoagulants in patients with atrial fibrillation; Eur Heart J. 2018 Apr 21;39(16):1330-1393.
5. J.D Douketis, J.M Anderson, D.M. Arnold; The Perioperative Anticoagulant Use for Surgery Evaluation (PAUSE) Study for Patients on a Direct Oral Anticoagulant Who Need an Elective Surgery or Procedure: Design and Rationale; Thromb Haemost. 2017 Dec;117(12):2415-2424.
6. NSW Government; Guidelines on Perioperative Management of Anticoagulant and Antiplatelet Agents; 2018
7. Northern Devon Healthcare NHS; Perioperative Anticoagulation and Antiplatelet Guideline; 2019
8. Handbook of Perioperative Medicine UKCPA 2016
9. C.L Garwood, J.S. Gourtney, T.L. Corbett et al; Is there a role for fondaparinux in perioperative bridging? Am J Health Syst Pharm. 2011 Jan 1;68(1):36-42.
10. M.Vaglimigli, H. Bueno, R.A. Byrne et al; 2017 ESC focused update on dual antiplatelet therapy in coronary artery disease developed in collaboration with EACTS: The Task Force for dual antiplatelet therapy in coronary artery disease of the European Society of Cardiology (ESC) and of the European Association for Cardio-Thoracic Surgery (EACTS); *European Heart Journal*, Volume 39, Issue 3, 14 January 2018, Pages 213–260
11. M. Moster, D. Bolliger; Perioperative Guidelines on Antiplatelet and Anticoagulant Agents: 2022 Update; BLOOD MANAGEMENT (KA TANAKA, SECTION EDITOR) [Published: 19 January 2022](#)
12. P.G Chassot, C. Marcucci, A.Delabays et al; Perioperative antiplatelet therapy; Am Fam Physician, 2010 Dec 15;82(12):1484-9.
13. J.D. Douketis, R.C Becker, J.A. Caprini et al; Perioperative Bridging Anticoagulation in Patients with Atrial Fibrillation; N Engl J Med 2015; 373:823-833
14. Heuts, S., Ceulemans, A., Kuiper, G. J., Schreiber, J., Van Varik, B. J., Olie, R. H., Cate, H. T., Maessen, J. G., Milojevic, M., & Maesen, B. (2023). Optimal management of cardiac surgery patients using direct oral anticoagulants: recommendations for clinical practice. *European Journal of Cardio-Thoracic Surgery*, 64(4). <https://doi.org/10.1093/ejcts/ezad340>
15. Dimitrova, G., Tulman, D. B., & Bergese, S. D. (2012). Perioperative management of antiplatelet therapy in patients with drug-eluting stents. HSR proceedings in intensive care & cardiovascular anesthesia, 4(3), 153–167.
16. Task Force on Patient Blood Management for Adult Cardiac Surgery of the European Association for Cardio-Thoracic Surgery (EACTS) and the European Association of Cardiothoracic Anaesthesiology (EACTA), Boer, C., Meesters, M. I., Milojevic, M., Benedetto, U., Bolliger, D., von Heymann, C., Jepsson, A., Koster, A., Osnabrugge, R. L., Ranucci, M., Ravn, H. B., Vonk, A. B. A., Wahba, A., & Pagano, D. (2018). 2017 EACTS/EACTA Guidelines on patient blood management for adult cardiac surgery. *Journal of cardiothoracic and vascular anesthesia*, 32(1), 88–120. <https://doi.org/10.1053/j.jvca.2017.06.026>
17. Douketis JD, Berger PB, Dunn AS, et al. The perioperative management of antithrombotic therapy: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines (8th Edition)., Chest, 2008, vol. 133 6 suppl(pg. 299S-339S)